CPUs can easily hit 5.2 and 5.3 GHz on air cooling with the voltage set to 1.35-1.36v. With AMD, the headroom is a bit limited and the lower-end SKUs have slightly greater headroom while the high-end SKUs are very tightly OCed from the factory. You can even go higher with the voltage that what we've mentioned but you should know that you will be reducing the life of your CPU.

THE BASICS

The simplest way to overclock your CPU would be to make use of the performance profiles on your motherboard or the accompanying software. All motherboards that support overclocking will have a dedicated section for increasing the CPU multiplier. If your processor is rated for, say 4.0 GHz, then you will see the multiplier set to x40. Depending on whether you are using an Intel CPU or an AMD CPU, there will be safe multiplier values that you can try without worry. Just ensure that the temperatures are well below 80 degrees celsius before you begin, otherwise you will end up triggering the thermal throttling and end up with lower performance than what you started with.

If not the BIOS, you can always use Intel's Extreme Tuner Utility (XTU) or AMD's Ryzen Master software to overclock. If you're on XTU, open the panel labelled 'Basic Tuning' and increase the Core Multiplier by 1x and then run the Benchmark. If Windows doesn't crash, then you're alright and can even try increasing the multiplier once again by 1x, and so on. On AMD Ryzen Master, simply head over to 'Profile 1' in the profile bar at the bottom and bump the clock speed of all cores by 25 MHz and run any benchmark. Keep doing this till it crashes and then revert to the stable value.

OVERCLOCKING - INTEL

Now that you've understood the basics and have your rig set up with the right components, let's delve into the details. Depending on your motherboard manufacturer, the labels for the different BIOS settings will be different. Start off by increasing the CPU multiplier to what's deemed safe for your CPU. If you have a new Coffee Lake Core i7-8700K, then you should start with x49 since practically all of them can easily hit 4.9 GHz on air cooling. Also, increase the FCLK to 1GHz which will help with increase the GPU performance and does not affect your BCLK. By default keep all settings on 'Auto' unless explicitly asked to change them. Modern motherboard BIOS can easily compensate for minor voltage and timing variations without requiring manual intervention. If you have a setting to enable or disable AVX then you'll have to set the AVX offset as well. AVX consumes a lot of power so applications that use AVX will destabilise your overclock if you haven't set the offset. For example, if your offset is a multiplier of -2x and the CPU is overclocked to 5.2 GHz, then the CPU will run at 5.2GHz until an application that uses AVX is run. When this happens, the CPU will drop to 5.0GHz [x52(multiplier)-x2(AVX offset)] to ensure stability.

Then you'll have to set the CPU cache/uncore/ring ratio to either match the CPU multiplier or remain within the vicinity of the CPU multiplier. Most of the time, your cache/uncore/ring ratio will be about 4-5 multipliers below the CPU multiplier.

Before starting off with the VCore, it's advisable to to check the Speed Step, Speed Shift and C-state settings. Keep them turned on since they actually help with the overclocking algorithm. You can disable C-states to keep the CPU running at peak power all the time but that wouldn't be advisable. Only when performing competitive overclocking should you keep these settings disabled.

We'll then move onto changing the VCore to ensure stability. VCore is the main voltage rail for the CPU and the cache as well. So whenever you increase VCore, both regions of the CPU will receive the extra juice. Ideally, you should start with the increasing the main CPU multiplier till you hit a stable overclock, then increase the cache/uncore/ring ratio till stable and then increase the VCore. There's also an additional setting called VID which is basically a voltage offset that kicks in based on the multiplier. Intel specifications call for a reduction in VCore when the multiplier and load are increased. However, when this happens, your overclock will become unstable. In order to compensate for this, there's an additional parameter called Load Line Calibration which reverses this voltage drop and ensure stability. Ensure that your voltage levels remain within the acceptable range for your CPU which would be 1.25-1.35v for more Intel CPUs.

At this point, we should boot into the system and run a benchmark to check for stability. Super Pi is a good one for this purpose. If everything runs stable, go through the steps again and increase the multipliers before increasing the VCore to achieve higher overclocks. The entire process is a tedious one and it'll require a lot of patience.

OVERCLOCKING - AMD

The instructions for AMD CPUs are more or less the same. The general sequence involves increasing the multiplier followed by stability tests. If the overclock isn't stable, increase the voltage by a fraction and go back to increasing the multiplier. With AMD boards, the underlying interconnect is linked with the RAM speeds, so higher RAM speeds contribute to a hefty performance increase.

Start off by increasing the core multiplier and then you'll have to increase the VCore to add stability. The new Ryzen CPUs have a more granular multiplier of .25x so when you start nearing the upper limit of your multiplier, increase the multiplier in a granular fashion rather than going up by x1 with each attempt. VCore can be between 1.35 to 1.4v and can go up to 1.5v as well. However, it's best to stick to 1.35v for moderate overclocks and below 1.45v for a system that runs 24x7. Anything higher is best left for the competitive scene. Sometimes, disabling the multithreading feature (called SMT) helps with stability but it halves your logical thread count so the next performance increment might not be desirable. By default, the BIOS will bump the VCore as you keep increasing the multiplier, this might overvolt your CPU, so it's best to set this manu-